

EXERCISE 1' SQL FUNDAMENTALS - SELECT / FILTER

Q 1. Query: Retrieve all columns from the employees table

Query

SELECT *

FROM employees;

Output:

employee - columns

id	first_name	last_name	department	salary	hire_date	city
1	Alice	Green	IT	70000	2020-01-10	Johannesburg
2	Brian	Lee	HR	45000	2019-03-22	Cape Town
3	Cathy	Zulu	Finance	65000	2018-07-18	Durban
4	David	Mokoena	Marketing	50000	2021-11-05	Pretoria
5	Eva	Naidoo	IT	72000	2017-09-30	Johannesburg

Q2 Find all unique departments

Query:

SELECT DISTINCT department

FROM employees;

Output

distinct - department

department

IT

HR

Marketing

Finance

Q3 Retrieve first and last names ordered by salary descending

Query

```
SELECT first_name,  
       last_name  
FROM employees  
ORDER BY salary DESC;
```

Output

employees salary

first_name	last_name
Eva	Naidoo
Alice	Green
Cathy	Zulu
David	Mokoena
Brian	Lee

Q4 Retrieve the top 3 highest-paid employees

Query

```
SELECT id,  
       first_name,  
       last_name,  
       salary  
FROM employees  
ORDER BY salary DESC  
LIMIT 3;
```

Output:

top paid-employees

id	first_name	last_name	salary
5	Eva	Naidoo	72000
1	Alice	Green	70000
3	Cathy	Zulu	65000

Q5 Find employees in the IT department.

Query

```
SELECT id,  
        first_name,  
        last_name,  
        department  
FROM employees  
WHERE department = 'IT';
```

Output

IT_department_employees

id	first_name	last_name	department
1	Alice	Green	IT
5	Eva	Naidoo	IT

Q6 Find employees in Finance with salary > 60000

Query

```
SELECT id,  
        first_name,  
        last_name,  
        department,  
        salary
```

FROM employees

WHERE department = 'Finance'

AND salary > 60000;

Output

Finance_salaries

id	first_name	last_name	department	salary
3	Cathy	Zulu	Finance	65000

07 Find employees in HR or Marketing

Query

SELECT id,

first_name,

last_name,

department

FROM employees

WHERE department IN ('HR', 'Marketing');

Outcome

employee_departments			
id	first_name	last_name	department
2	Brian	Lee	HR
4	David	Mokoena	Marketing

Q8 Find employees not in IT

Query id,

first_name,

last_name,

department

FROM employees

WHERE NOT department = 'IT';

Outcome

employees_not_IT			
id	first_name	last_name	department
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
4	David	Mokoeng	Marketing

Q9 Find employees in IT, HR or Finance using IN.

Query

```
SELECT id,  
       first_name,  
       last_name,  
       department  
FROM employees  
WHERE department IN ('IT', 'HR', 'Finance');
```

Outcome

selected_departments

id	first_name	last_name	department
1	Alice	Green	IT
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
5	Eva	Naidoo	IT

Q10 Find employees in IT with salary > 65000 and city Johannesburg.

Query

```
SELECT *  
FROM employees  
WHERE department = 'IT'  
AND salary > 65000  
AND city = 'Johannesburg';
```

Outcome

Q10 Outcome

Id	first-name	last-name	department	salary	hire-date	city
1	Alice	Green	IT	70000	2020-01-10	Johannesburg
5	Eva	Naidoo	IT	72000	2017-09-03	Johannesburg